

Orchestrating Granular Access for Sensitive Omics Data: The SAFES Architecture within NCHC's Trusted Cloud Platform



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National Center for High-performance Computing

Founded in 1991, the National Center for High-performance Computing (NCHC) is dedicated to advancing Taiwan's national HPC capabilities and providing world-class supercomputing and research network infrastructure.

As the **nation's only** shared large-scale computing and research network facility, NCHC offers advanced HPC, AI, and data services that accelerate scientific discovery, drive technical innovation, and cultivate cross-disciplinary talent.

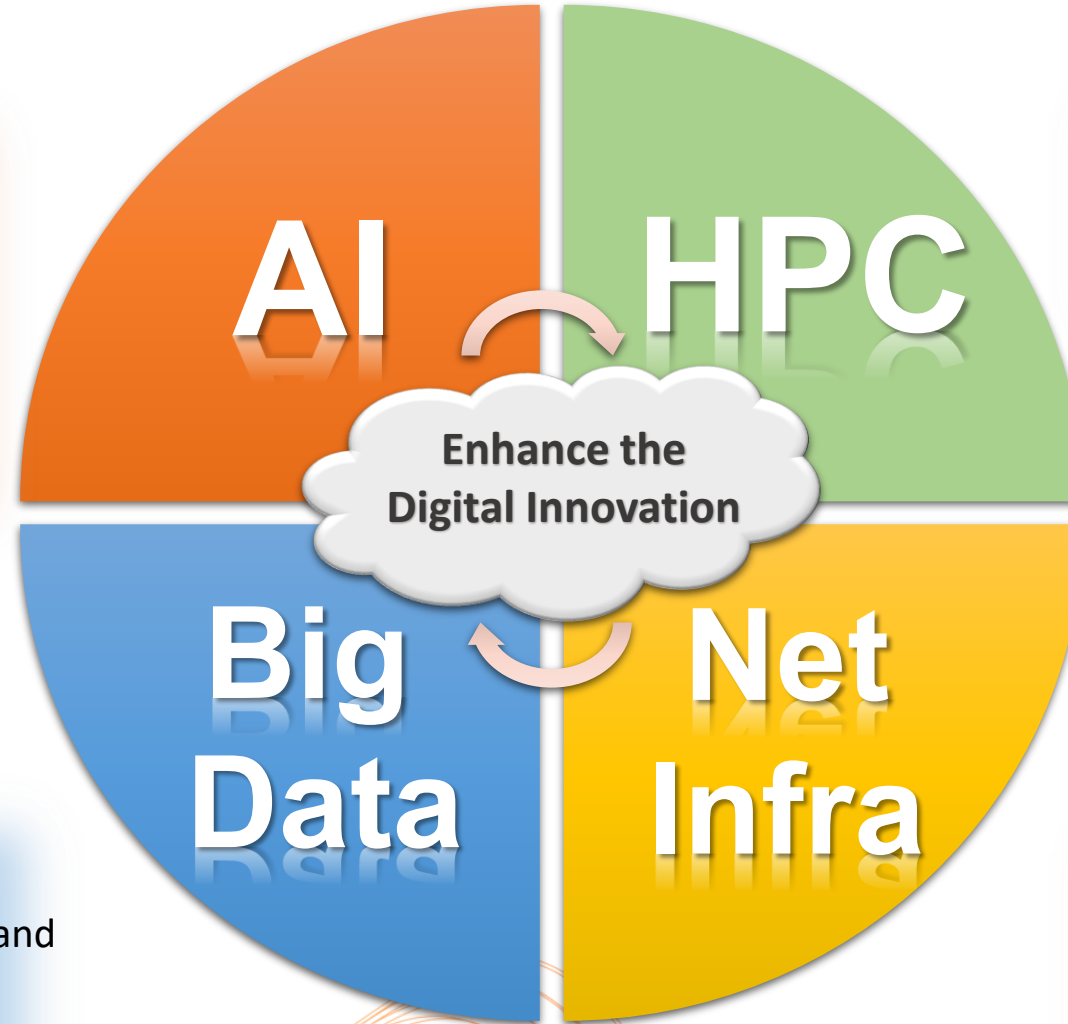
Furthermore, NCHC enhances **high-accuracy simulation, trusted data platforms, generative AI**, and **digital twin** integration to foster cross-domain innovation, while pioneering **quantum computing** to shape a smarter and more sustainable future.



4 Major Mission Pillars

- **Deploy** national long-term large-scale AI computing power
- **Build** a Generative AI (GenAI) model service platform
- **Accelerate** GenAI model deployment and practical applications
- **Drive** Taiwan's GenAI R&D and international integration

- **Establish** a multi-domain big data specialty cloud platform
- **Deliver** a highly trusted environment and dedicated cloud services for sensitive data
- **Unlock** innovative value through data reuse



- **Promote** key national HPC initiatives
- **Provide** a high-quality, reliable R&D computing environment
- **Cultivate** high-performance computing talent

- **Construct** national-level broadband network infrastructure
- **Expand** cloud data center capacity in Southern Taiwan
- **Strengthen** national network resilience

Create a Cloud Big Data
Computing Ecosystem

3 Major Developments Goals

National-Level Data
Services

Digital Governance
Services

Cross-Domain Integrated
Application Services

Intelligent
Tools

Multi-
Platform

Customized
Services

Testing &
Experimentation
Field

**Secure Cloud Data
Analytics Infrastructure
(Trusted-Cloud Platform)**

Zero Trust Network
Architecture

Data Protection
Mechanism

Network Proxy
Encryption Mechanism

Stateless Computing
Mechanism

Infra



Supercomputers



VM



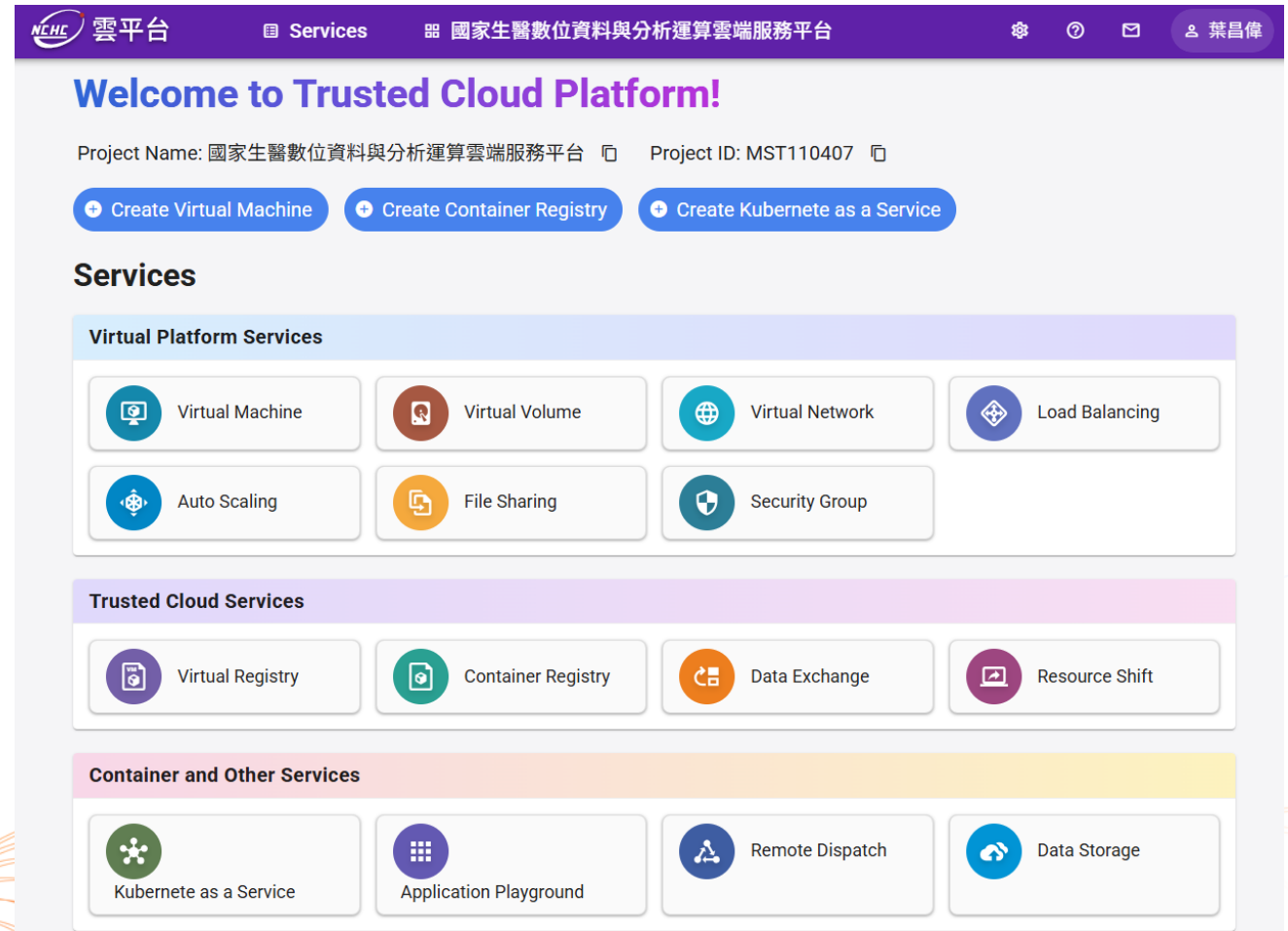
Fiber Network



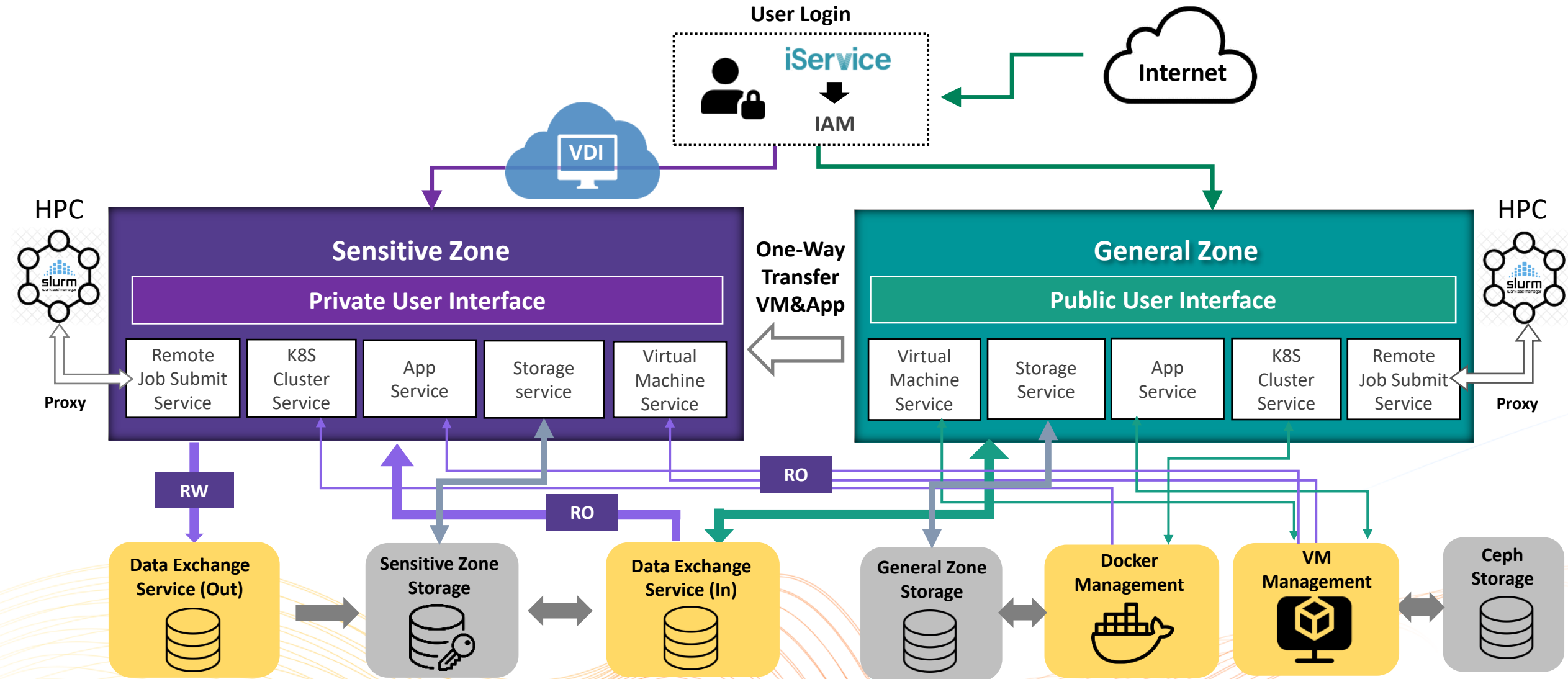
Cloud Compute Centers

Trusted-Cloud Platform (TCP)

- **Secure and reliable Cloud environment**
 - Designed with advanced security controls and storage encryption
 - Ensures strong data protection and compliance
- **Flexible computing services**
 - Offers **Virtual Machine (VM)** services
 - Provides **Kubernetes-based container** services
 - Meets diverse user needs with flexibility
- **Value-added intelligent service**
 - Includes the **Trusted-Cloud Intelligent Execution Assistant and QA Assistant**
 - Simplifies Cloud operations
 - Enhances overall user productivity



Trusted-Cloud Platform (TCP) Architecture



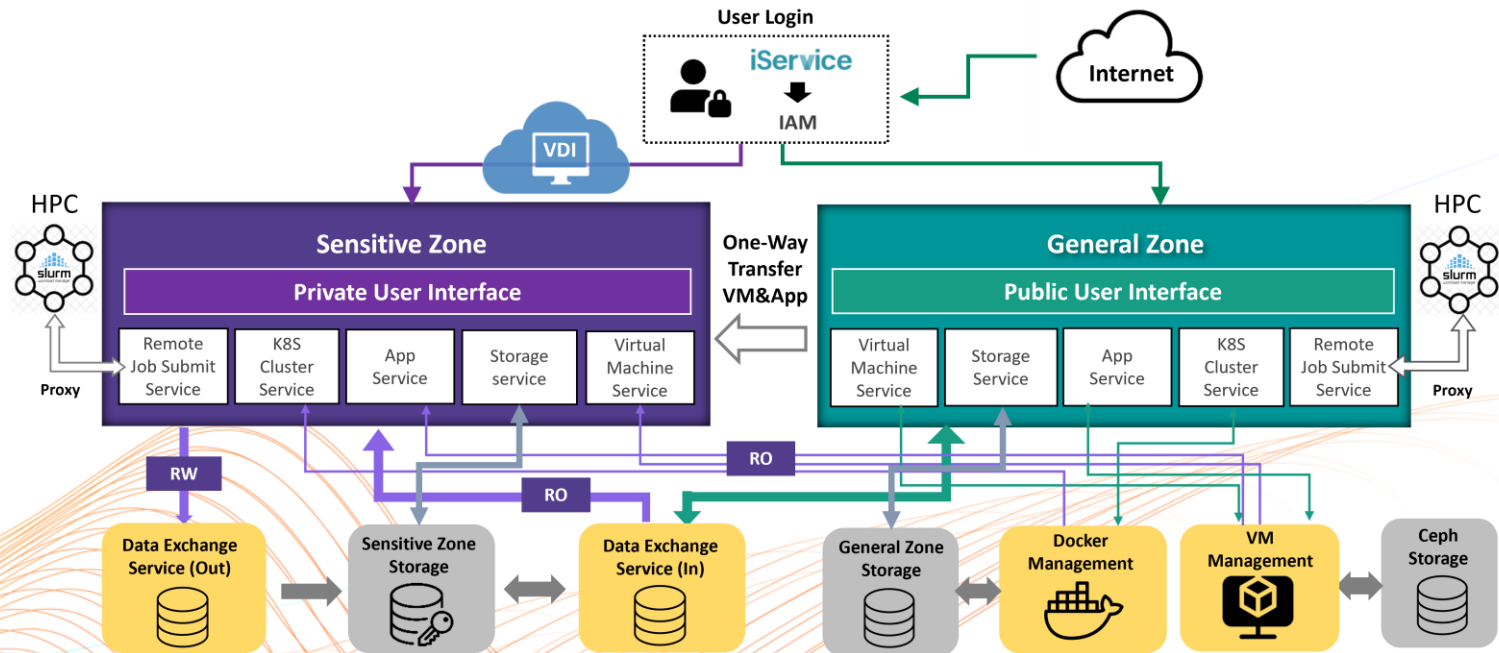
Trusted-Cloud Platform (TCP) Architecture

The TCP is a partitioned platform which is divided into two zones:

1. **General Zone (Public Zone)** : Designated for general-purpose applications.
2. **Sensitive Zone (Private Zone)** : Dedicated to handling sensitive data.

Cross-zone data sharing is controlled by the **Data Exchange Service**, underpinned by a robust infrastructure of multiple storage, application, and computing services.

Nevertheless, TCP may not fully meet all the usage scenarios and stringent compliance and security requirements for sensitive big data services.

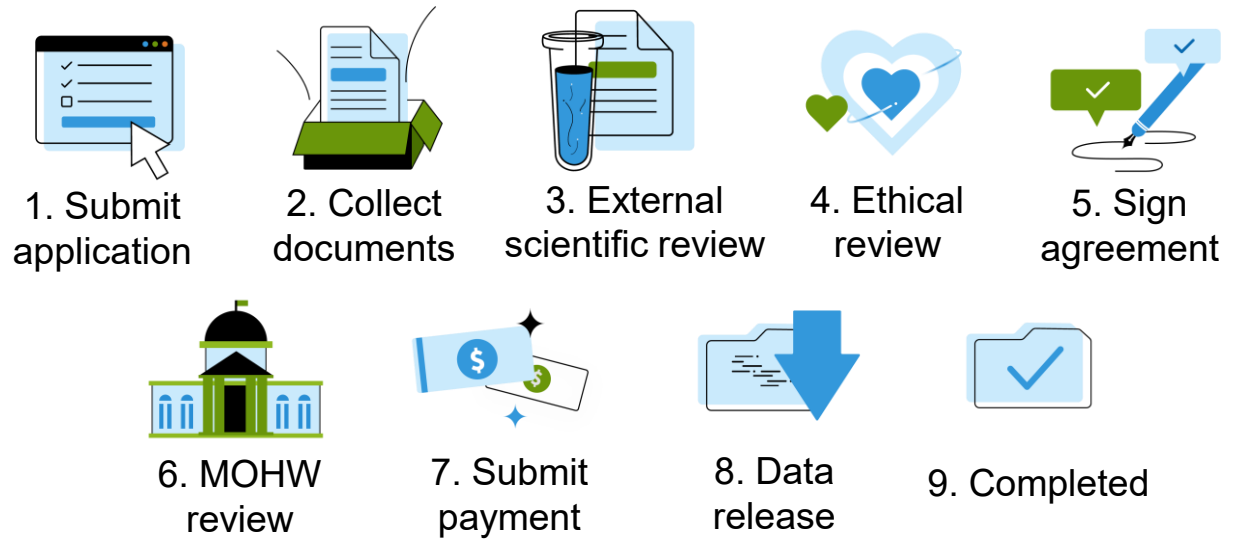


Enduring Partnership: NCHC and Taiwan Biobank

~2,000 Whole Genome Sequencing, alignment, variant calling and ~200,000 genotyping data.

Easing the Burden on IT

- **1.3+ PB Secure Storage:** Protected by the 3-2-1 backup rule
- **High-Availability (HA) Service:** Ensuring maximum uptime
- **24/7 Monitoring:** Compliant with ISO 27001 standards
- **Daily Updates:** Keeping data continuously synchronized
- **Flexible Access:** Available via authorized web services and HPC environments



Taiwan Cancer Moonshot Project

Strategic Core & Data

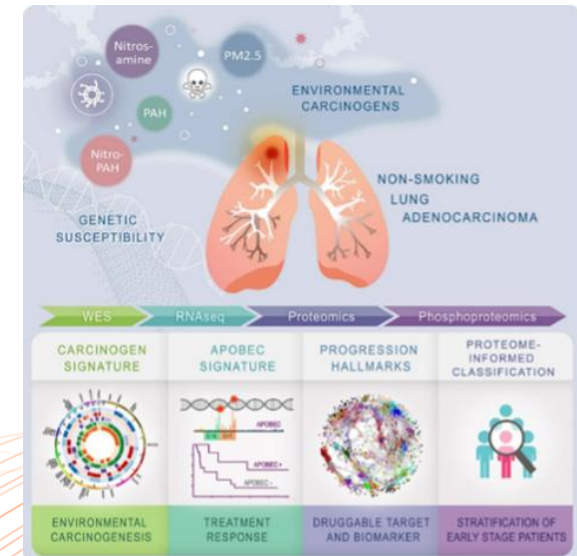
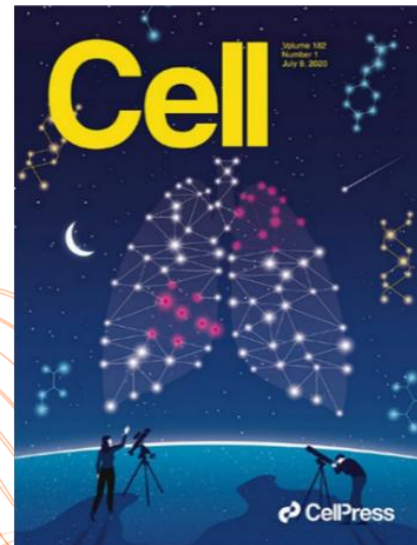
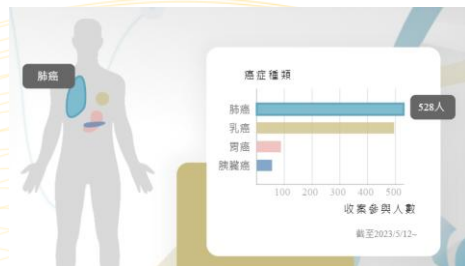
- **Elite Integration:** Synergizes top-tier teams from Academia Sinica, NTU, and major medical centers.
- **Pioneering Multi-Omics:** Decodes biology by combining genomics, transcriptomics, proteomics, phosphoproteomics, and clinical records.
- **Overarching Goal:** Uncovers patient molecular heterogeneity to discover actionable precision medicine solutions.

Landmark Breakthroughs

- **2020 Cell Cover:** Unveiled the world's first deep multi-omics map of non-smoking lung cancer in East Asian cohorts.
- **2025 Cancer Cell Study:** Conducted the first large-scale lung adenocarcinoma proteogenomic study bridging Asian and Western cohorts.

National Knowledgebase

- Built in collaboration with the National Center for High-Performance Computing (NCHC).
- **Dynamic Portal:** Offers dynamic interactive visualizations, expanding from lung cancer to breast cancer.
- **Secure Framework:** Operates under a "data-stay-in-place" protocol to accelerate local clinical drug discovery.

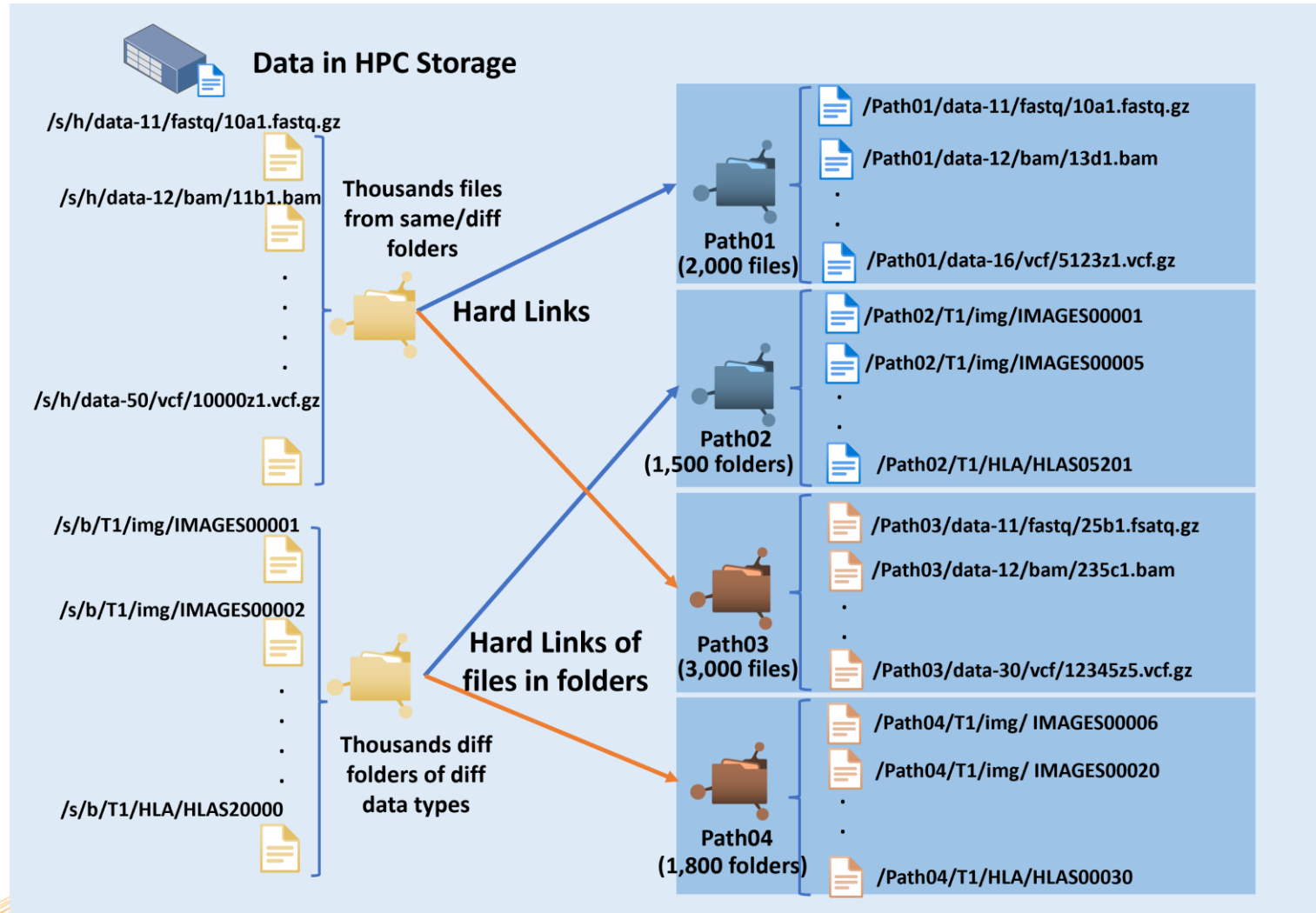


Data Privileges Assignment in HPC

Data Access & Management Policies in HPC Storage.

- Granular Authorization: Access is granted strictly to approved subsets - either at the folder level or file level.
- Storage Optimization: Leveraging Hard Links and File ACL for permission management to eliminate data duplication.
- Zero-Copy Efficiency: Ensures secure data sharing without consuming extra HPC storage capacity.

Data Loss Prevention at Endpoints?
Cloud Platform?





Sensitive-data Access, File-Exchange System (SAFES)

Architectural Decoupling of **SAFES**:

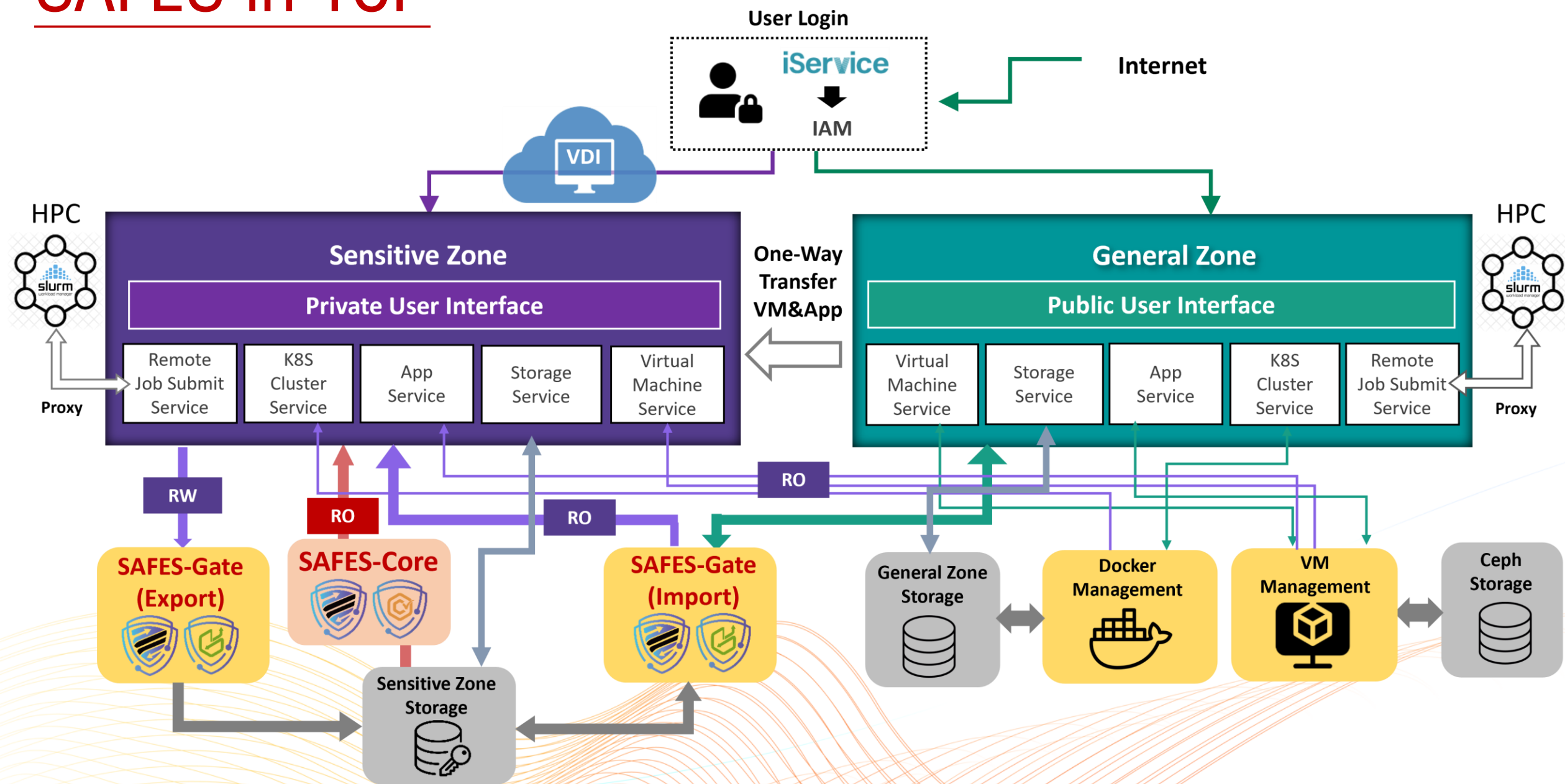


SAFES-Core: Manages the authorization logic.



SAFES-Gate: Handles the actual data exchange mechanics.

SAFES in TCP



iRODS' roles in SAFES

iRODS®



Data Virtualization

Data Discovery

Workflow Automation

Secure Collaboration

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Data Virtualization: one of four Core Competencies of iRODS

- Access Paradigm
 - Data stored within the system is typically accessed exclusively through an iRODS Client.
- User View
 - Files are presented as Data Objects abstractly organized into Collections. Collections are completely decoupled from infrastructure, making no reference to actual physical storage paths.
- Storage View
 - Location Transparency: Two Data Objects within the exact same Collection can reside in entirely different physical storage locations or hardware.

I can register and combine existing physical files in NCHC's storage into the iCAT catalog without moving or copying it.

iRODS' roles in SAFES

iRODS®



Data Virtualization

Data Discovery

Workflow Automation

Secure Collaboration

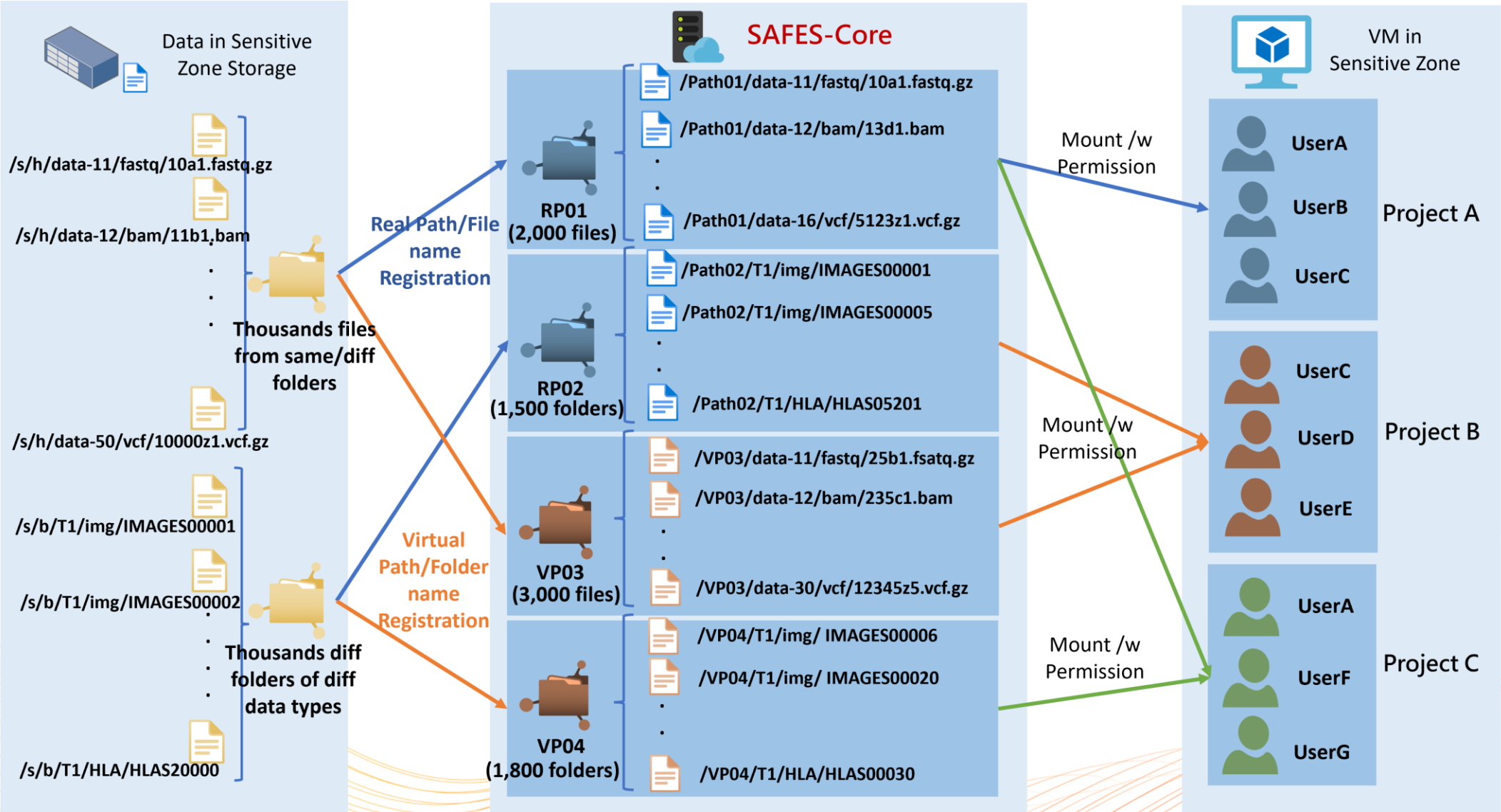
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Secure Collaboration: another Core Competencies of iRODS

- Collaboration Challenges
 - Secure dataset sharing across multiple workgroups is often essential.
 - As datasets scale beyond several petabytes, moving massive data volumes between physical locations becomes highly impractical or technically impossible.
- Access Control
 - Permissions are modeled after standard UNIX file systems, ensuring a familiar and robust security framework.
- Administrative Governance
 - Group memberships and roles are strictly defined and governed by system administrators to maintain organizational compliance.

I can also manage the permissions of data in iCAT catalog.

Data Privileges Assignment in SAFES-Core



Usage Scenario of SAFES-Core



Storage Admin

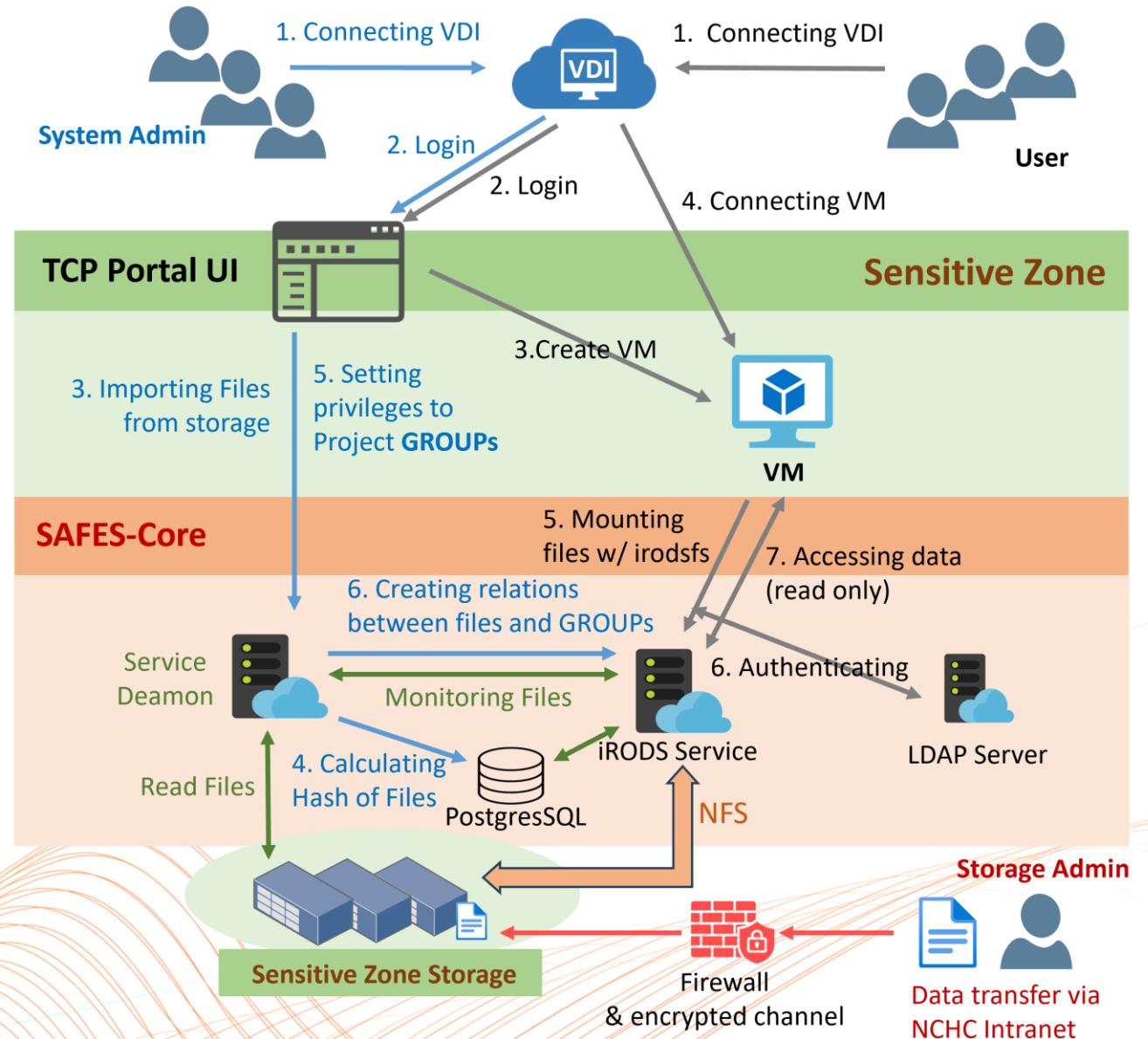
- Manages storage infrastructure and original data

System Admin

- Manages data access privileges authorized by Data Owner or Data Admin

User

- Accesses and utilizes data under TCP and SAFES-Core governance



Usage Scenario of SAFES-Gate

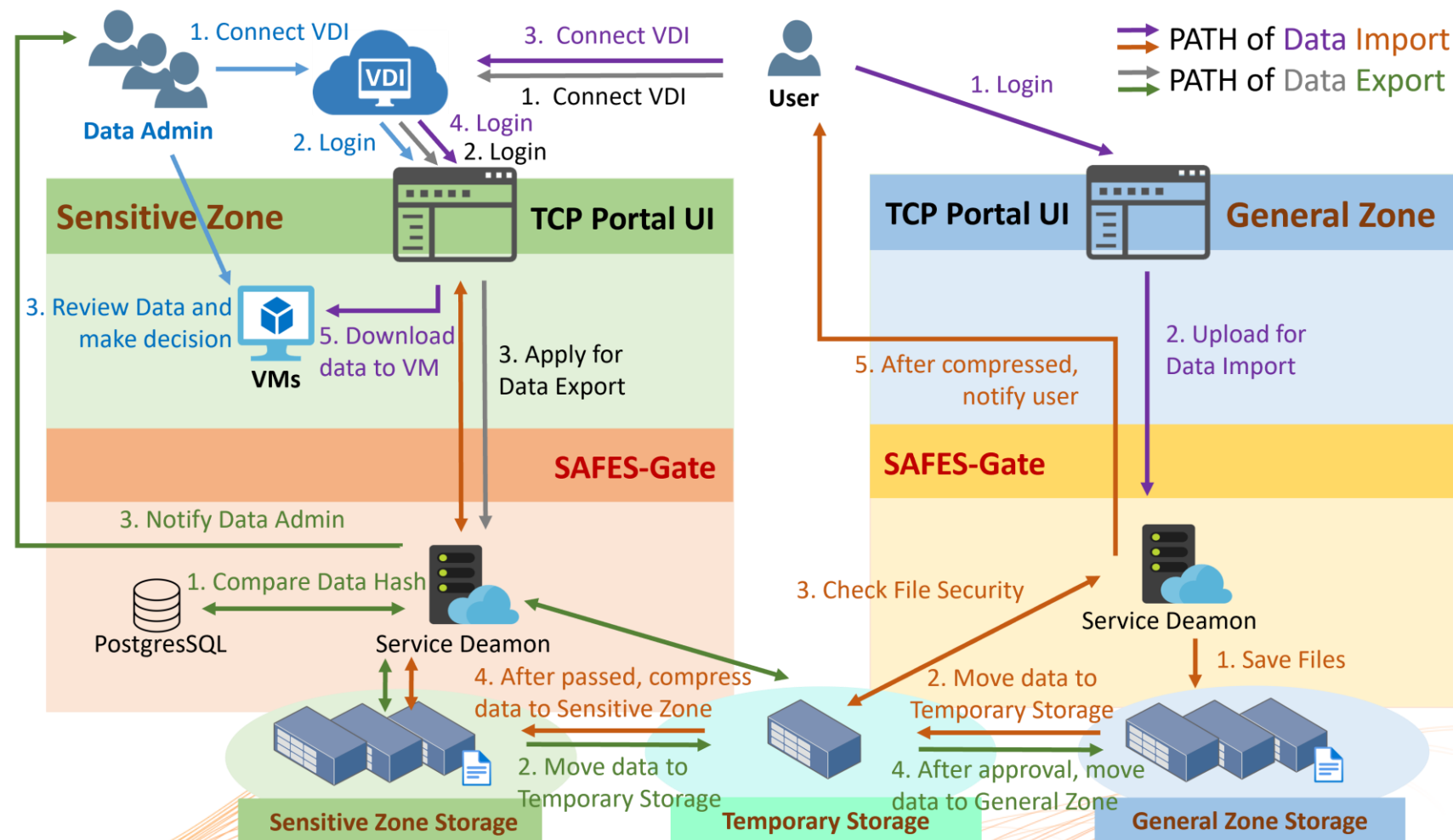


Data Admin

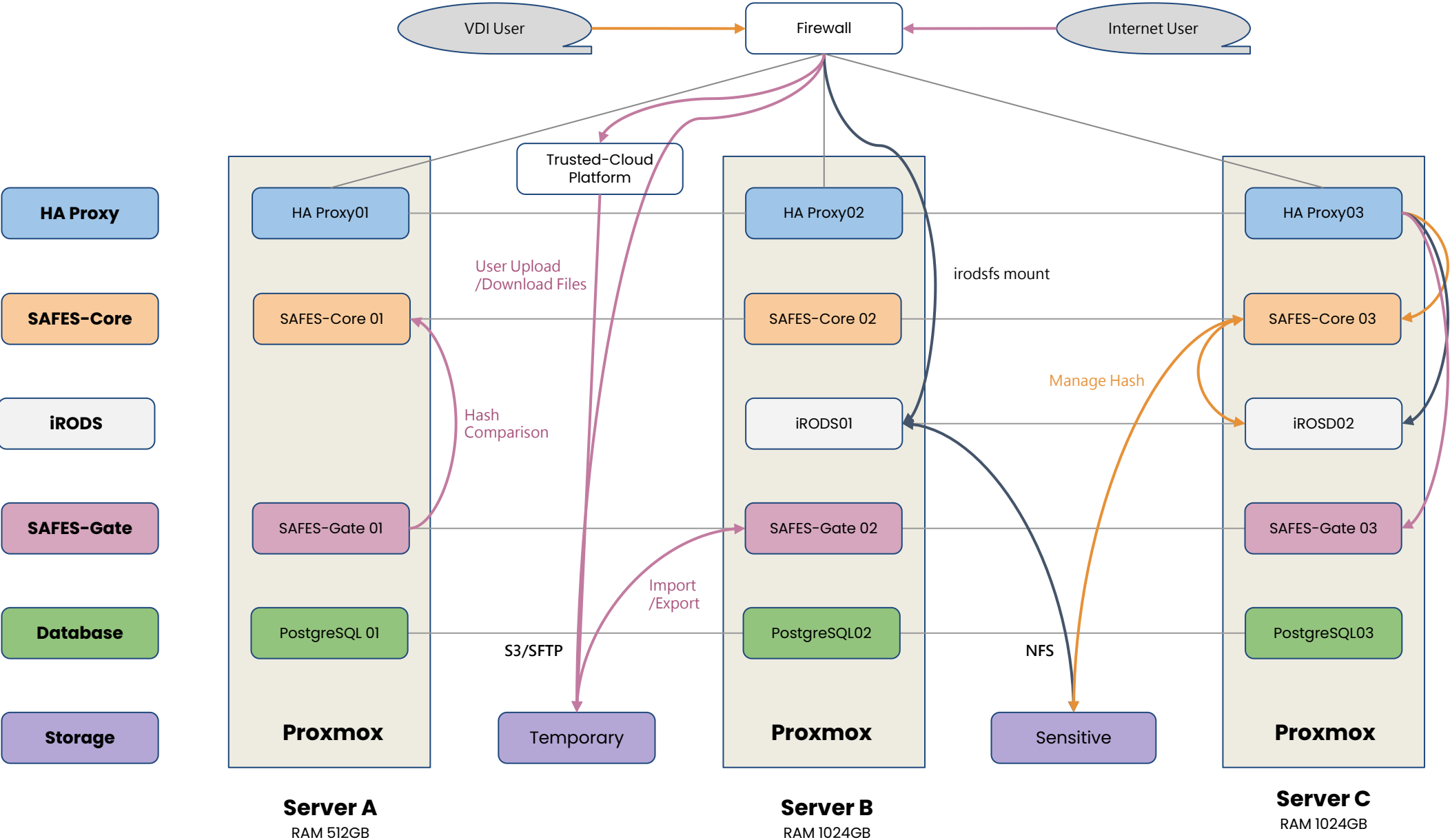
- Review and gatekeep data export

User

- Import or Export data under TCP and SAFES-Gate governance



System HA Architecture of SAFES



Conclusions

TCP & SAFES meet the **Five Safes Framework** of Trusted Research Environment

The Five Safes framework is a set of principles which enable data services to provide safe research access to data. The framework originated from the UK's Office for National Statistics (ONS) and was developed by them and other data providers in the 2010s.

- **Safe data**: data is treated to protect any confidentiality concerns.
- **Safe projects**: research projects are approved by data owners for the public good.
- **Safe people**: researchers are trained and authorised to use data safely.
- **Safe settings**: a SecureLab environment prevents unauthorised use.
- **Safe outputs**: screened and approved outputs that are non-disclosive.



Questions ?



Thank you